

Xiaoshan Wu

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Research Portfolio

Google Scholar



EDUCATION

The University of Hong Kong (HKU)

Ph.D. in Electrical and Electronic Engineering (Focus: 3D/4D Vision, World Models & Embodied AI)

Hong Kong
Sep. 2023 – Present

- **Awards:** Hong Kong PhD Fellowship (HKPFS, Top Tier), HKU Presidential Scholarship.
- **Advisor:** Prof. Xiaojuan Qi. **GPA:** 4.0/4.3.

Zhejiang University (ZJU) & UIUC

Dual B.S. in Computer Engineering (ZJUI Institute)

China & USA
Sep. 2019 – Jun. 2023

- **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

Self-Improving Physical Agents through Agent–World Model Co-Evolution. I study agents that turn physical interaction into cumulative improvement, with a focus on action-conditioned 3D/4D world models, embodied spatial reasoning, and closed-loop robot learning for robust planning and control.

PUBLICATIONS & PREPRINTS

* indicates equal contribution; † denotes project lead.

1. **X. Wu**, et al. **Verifiable 4D World–Action Model for Robust Robot Control.**
Under Review at ICLR 2027.
2. **X. Wu**, et al. **Game: Geometry as Memory for Multi-View Spatial Reasoning.**
Under Review at ICRA 2027.
3. T. Wan*, **X. Wu***†, Y. Yu, et al. **LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.**
Under Review at IEEE TPAMI.
4. M. Liu*, X. Lyu*, T. Ren, P. Dai, **X. Wu**, et al. **FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.**
ECCV 2026.
5. Y. Yu*, **X. Wu***, X. Hu, et al. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.**
ECCV 2026.
6. **X. Wu**, X. Lyu, Y. Yu, et al. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.**
ICLR 2026.
7. X. Lyu, M. Liu, **X. Wu**, et al. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
CVPR 2026.
8. Y. Yu, X. Zhang, S. Wang, ..., **X. Wu**, et al. **Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.**
Nature, 2026.
9. **X. Wu***, Y. Yu*, X. Lyu, et al. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.**
NeurIPS 2025 (**Spotlight, Top 3.18%**).
10. B. Wang, X. Zhang, S. Wang, ..., **X. Wu**, et al. **Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.**
Science Advances, 2025.
11. H. Chen, J. Yang, J. Chen, ..., **X. Wu**, et al. **Continuous-Time Digital Twin with Analog Memristive Neural ODE Solver.**
Science Advances, 2025.
12. C. Li, X. Chen, Z. Zhang, **X. Wu**, et al. **Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.**
Nano Letters, 2024.
13. Y. Yu, S. Wang, M. Xu, ..., **X. Wu**, et al. **Random Memristor-Based Dynamic Graph CNN for Efficient Point Cloud Learning at the Edge.**
npj Unconventional Computing, 2024.
14. **X. Wu**, W. He, M. Yao, et al. **Event-Based Depth Prediction with Deep Spiking Neural Network.**
IEEE Transactions on Cognitive and Developmental Systems (TCDS), 2024.

RESEARCH EXPERIENCE

Harvard University (SEAS)

Visiting Researcher

Cambridge, MA, USA

Apr. 2026 – Aug. 2026

- Collaborated with Yilun Du and Mengyu Wang on self-improving physical agents and agent–world model co-evolution.

TECHNICAL SKILLS

Research Expertise: Self-Improving Physical Agents, World–Action Models, 3D/4D Vision, Spatial Reasoning, Video Generation, Event Vision.

Frameworks & Tools: PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS)

2023

HKU Presidential Scholarship

2023

Outstanding Graduate of Zhejiang University

2023